

8-Bit Buffers with Schmitt Trigger Inputs

SN54/74LS310 SN54/74S310
SN54/74LS340 SN54/74S340
SN54/74LS341 SN54/74S341
SN54/74LS344 SN54/74S344

Features

- Schmitt-trigger inputs guarantee high noise margin
- Three-state outputs drive bus lines
- Typical input and output capacitance ≤ 10 pf
- Low-current PNP inputs reduce loading
- 20-pin SKINNYDIP® saves space
- 8-bit data path matches byte boundaries
- Ideal for microprocessor interface
- Complementary-enable '310 and '341 types combine multiplexer and driver functions
- Pin-compatible with SN54/74S210/240/1/4 and SN54/74LS210/240/1/4; can be direct replacement in systems with noise problems

Description

In addition to the standard Schottky and low-power Schottky 8-bit buffers, Monolithic Memories provides full hysteresis with a "true" Schmitt-trigger circuit. The improved performance characteristics are designed (1) for the low-power Schottky buffers, to be consistent with the SN54/74LS14 hex Schmitt-trigger inverter, and to guarantee a full 400 mV noise immunity; (2) for the Schottky buffers, to have low propagation delays, and to guarantee a full 500 mV noise immunity. The Schmitt-trigger operation makes these LS/S buffers ideal for bus receivers in a noisy environment.

These 8-bit buffers provide high-speed and high-current interface capability for bus-organized digital systems. The three-state drivers will source a termination to ground (up to 133 Ω) or sink a pull-up to V_{CC} as in the popular 220 Ω /330 Ω computer

Ordering Information

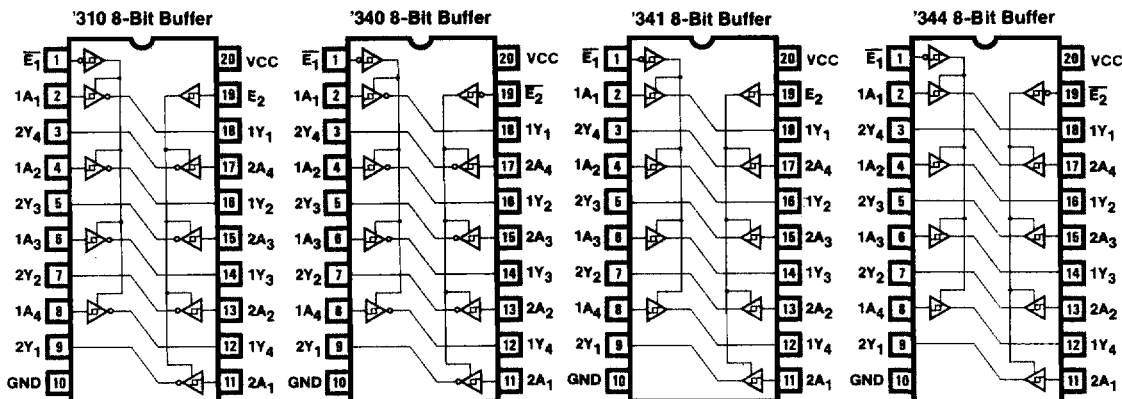
PART NUMBER	PKG*	TEMP	ENABLE	POLARITY	POWER
SN54LS310	J,F	mil	High- Low	Invert	LS
SN74LS310	N,J	com			
SN54LS340	J,F	mil	Low		
SN74LS340	N,J	com			
SN54LS341	J,F	mil	High- Low	Non- Invert	
SN74LS341	N,J	com			
SN54LS344	J,F	mil	Low		
SN74LS344	N,J	com			
SN54S310	J,F	mil	High- Low	Invert	S
SN74S310	N,J	com			
SN54S340	J,F	mil	Low		
SN74S340	N,J	com			
SN54S341	J,F	mil	High- Low	Non- Invert	
SN74S341	N,J	com			
SN54S344	J,F	mil	Low		
SN74S344	N,J	com			

peripheral termination. The PNP inputs provide improved fan-in with 0.2 mA I_{IL} for the low-power Schottky buffers and 0.25 mA I_{IL} for the Schottky buffers.

The '340 and '344 provide inverting and non-inverting outputs respectively, with assertive-low enables. The '310 and '341 also provide inverting and non-inverting outputs respectively, but with complementary (both assertive-low and assertive-high) enables, to allow transceiver or multiplexer operation.

All of the 8-bit devices are packaged in the popular 20-pin SKINNYDIP®.

Logic Symbols



*For other package types, please contact your local sales representative.

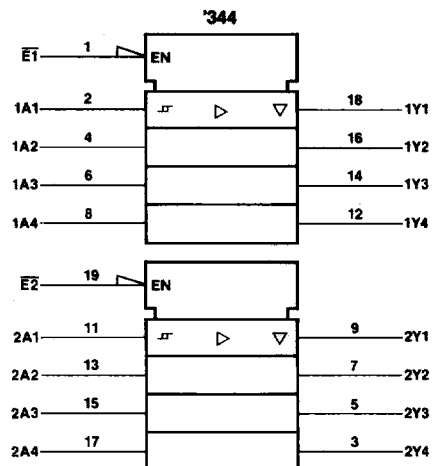
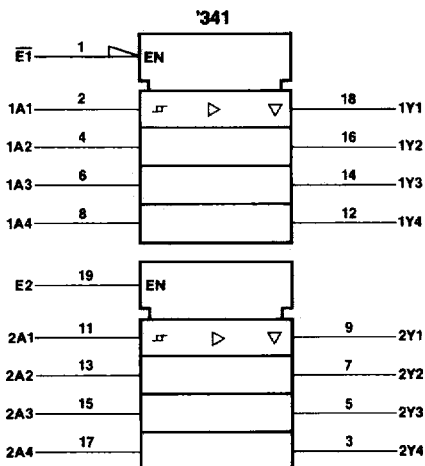
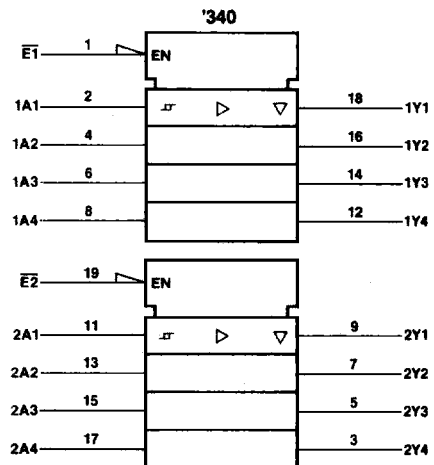
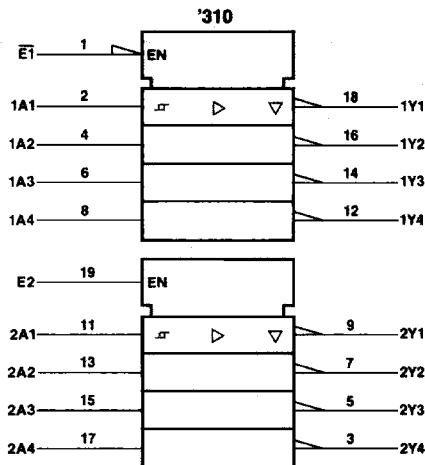
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Monolithic Memories

IEEE Symbols



Absolute Maximum Ratings

Supply voltage V_{CC}	7 V
Input voltage	7 V
Off-state output voltage	5.5 V
Storage temperature	-65° to +150° C

Operating Conditions

SYMBOL	PARAMETER	MILITARY			COMMERCIAL			UNIT
		MIN	TYP	MAX	MIN	TYP	MAX	
V_{CC}	Supply voltage	4.5	5	5.5	4.75	5	5.25	V
T_A	Operating free-air temperature	-55		125	0		75	°C

Electrical Characteristics Over Operating Conditions

SYMBOL	PARAMETER		TEST CONDITIONS		MILITARY MIN TYP MAX			COMMERCIAL MIN TYP MAX			UNIT
V _{T+}	Positive threshold voltage		Any A*		1.5	1.7	2.0	1.5	1.7	2.0	V
V _{T-}	Negative threshold voltage		Any A*		0.6	0.9	1.1	0.6	0.9	1.1	V
V _{IC}	Input clamp voltage		V _{CC} = MIN I _I = -18 mA		-1.5			-1.5			V
ΔV _T	Hysteresis (V _{T+} -V _{T-})		Any A*		0.4	0.8		0.4	0.8		V
ΔV _{DB}	Dead band voltage		Any A*		0.4			0.4			V
V _{IL}	Input low voltage		Any E*		0.8			0.8			V
V _{IH}	Input high voltage		Any E*		2.0			2.0			V
I _{IL}	Low-level input current		V _{CC} = MAX V _I = 0.4 V		-0.2			-0.2			mA
I _{IH}	High-level input current		V _{CC} = MAX V _I = 2.7 V		20						μA
I _I	Maximum input current		V _{CC} = MAX V _I = 7 V		0.1			0.1			mA
V _{OL}	Low-level output voltage		V _{CC} = MIN V _{T+} = 2 V V _{T-} = 0.6 V	I _{OL} = 12 mA	0.4			0.4			V
				I _{OL} = 24 mA				0.5			
V _{OH}	High-level output voltage		V _{CC} = MIN V _{T+} = 2 V V _{T-} = 0.6 V	I _{OH} = -3 mA	2.4	3.4		2.4	3.4		V
				I _{OH} = -12 mA	2						
				I _{OH} = -15 mA			2				
I _{OZL}	Off-state output current		V _{CC} = MAX V _{T+} = 2 V V _{T-} = 0.6 V	V _O = 0.4 V	-20			-20			μA
V _O = 2.7 V				20			20			μA	
I _{OS}	Output short-circuit current**		V _{CC} = MAX		-40	-225		-40	-225		mA
I _{CC}	Supply Current	Outputs High	V _{CC} = MAX Outputs open	'LS310,'LS340	17	27		17	27		mA
				'LS341,'LS344	18	35		18	35		
		Outputs Low		'LS310,'LS340	26	44		26	44		
				'LS341,'LS344	32	46		32	46		
		Outputs Disabled		'LS310,'LS340	29	50		29	50		
				'LS341,'LS344	34	54		34	54		

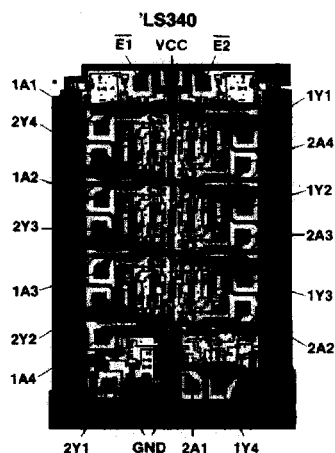
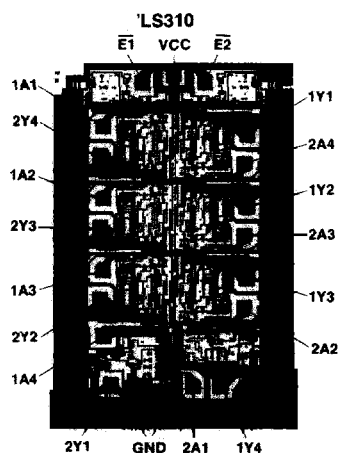
** Not more than one output should be shorted at a time and duration of the short-circuit should not exceed one second.

* "A" indicates data input, "E" indicates enable input.

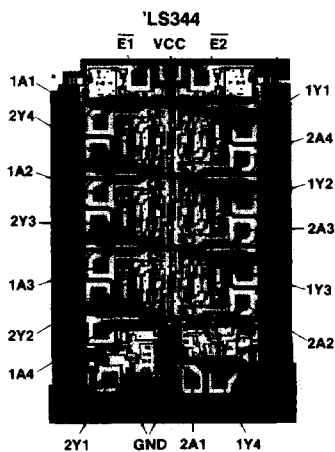
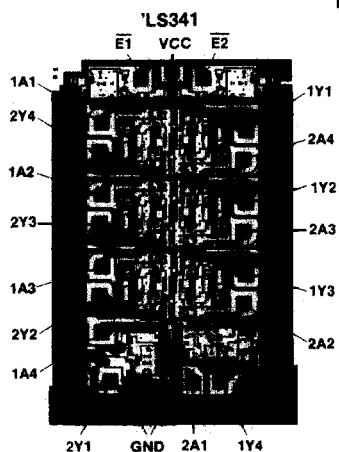
Switching Characteristics $V_{CC} = 5\text{ V}$, $T_A = 25^\circ\text{C}$

SYMBOL	PARAMETER	TEST CONDITIONS (See Test Load/Waveforms)	'LS310, 'LS340 MIN TYP MAX	'LS341, 'LS344 MIN TYP MAX	UNIT
t _{PLH}	Data to Output delay	C _L = 45 pF R _L = 667 Ω	19	25	ns
t _{PHL}			19	25	ns
t _{PZL}	Output Enable delay		32	40	ns
t _{PZH}			23	35	ns
t _{PLZ}	Output Disable delay	C _L = 5 pF R _L = 667 Ω	18	30	ns
t _{PHZ}			15	25	ns

Die Configurations



Die Size: 85x146 mil²



Absolute Maximum Ratings

Supply voltage V_{CC}	7.0 V
Input voltage	5.5 V
Off-state output voltage	5.5 V
Storage temperature	-65° to +150°C

Operating Conditions

SYMBOL	PARAMETER	MILITARY			COMMERCIAL			UNIT
		MIN	TYP	MAX	MIN	TYP	MAX	
V_{CC}	Supply voltage	4.5	5	5.5	4.75	5	5.25	V
T_A	Operating free-air temperature	-55		125	0		75	°C

Electrical Characteristics Over Operating Conditions

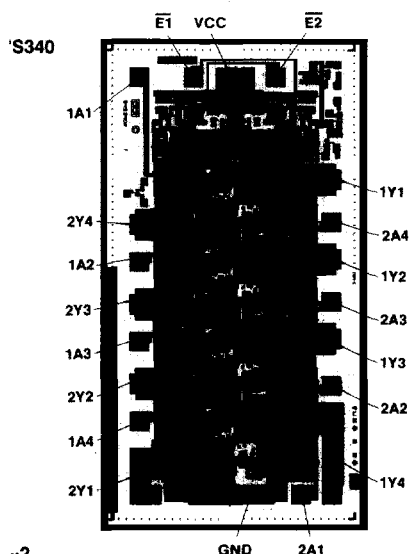
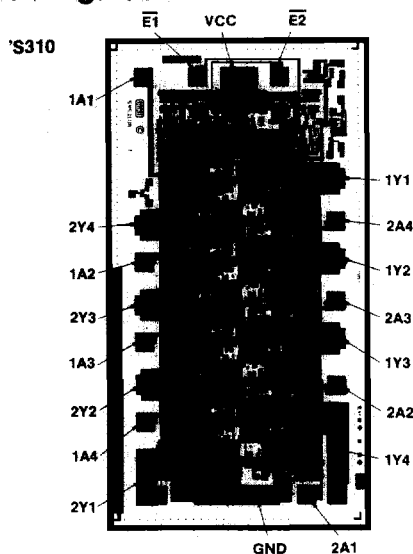
SYMBOL	PARAMETER	TEST CONDITIONS	MILITARY			COMMERCIAL			UNIT
			MIN	TYP	MAX	MIN	TYP	MAX	
V_{T+}	Positive threshold voltage	Any A*	1.5	1.8	2.05	1.6	1.8	2.0	V
V_{T-}	Negative threshold voltage	Any A*	0.8	1.1	1.35	0.8	1.1	1.3	V
V_{IC}	Input clamp voltage	$V_{CC} = \text{MIN}$ $I_I = -18 \text{ mA}$			-1.2			-1.2	V
ΔV_T	Hysteresis ($V_{T+} - V_{T-}$)	Any A*	0.5	0.7		0.5	0.7		V
ΔV_{DB}	Dead band voltage	Any A*	0.15			0.3			V
V_{IL}	Input low voltage	Any E*			0.8			0.8	V
V_{IH}	Input high voltage	Any E*	2.0			2.0			V
I_{IL}	Low-level input current	$V_{CC} = \text{MAX}$ $V_I = 0.5 \text{ V}$			-0.25			-0.25	mA
I_{IH}	High-level input current	$V_{CC} = \text{MAX}$ $V_I = 2.7 \text{ V}$			50			50	μA
I_I	Maximum input current	$V_{CC} = \text{MAX}$ $V_I = 5.5 \text{ V}$			1			1	mA
V_{OL}	Low-level output voltage	$V_{CC} = \text{MIN}$ $V_{T+} = 2 \text{ V}$ $V_{T-} = 0.8 \text{ V}$	$I_{OL} = 48 \text{ mA}$		0.55				V
			$I_{OL} = 64 \text{ mA}$			0.55			
V_{OH}	High-level output voltage	$V_{CC} = \text{MIN}$ $V_{T+} = 2 \text{ V}$ $V_{T-} = 0.8 \text{ V}$	$I_{OH} = -1 \text{ mA}$			2.7			V
			$I_{OH} = -3 \text{ mA}$		2.4 3.4	2.4 3.4			
			$I_{OH} = -12 \text{ mA}$		2				
			$I_{OH} = -15 \text{ mA}$			2			
I_{OZL}	Off-state output current	$V_{CC} = \text{MAX}$ $V_{IH} = 2.0 \text{ V}$ $V_{IL} = 0.8 \text{ V}$	$V_O = 0.5 \text{ V}$		-50			-50	μA
I_{OZH}			$V_O = 2.7 \text{ V}$		50			50	μA
I_{OS}	Output short-circuit current**	$V_{CC} = \text{MAX}$	-50		-225	-50		-225	mA
I_{CC}	Supply Current	Outputs High	$V_{CC} = \text{MAX}$ Outputs open	'S310, 'S340	50 80	50 80			mA
				'S341, 'S344	80 130	80 130			
				'S310, 'S340	110 155	100 155			
				'S341, 'S344	130 180	130 185			
				'S310, 'S340	135 180	135 180			
				'S341, 'S344	155 180	150 200			

** Not more than one output should be shorted at a time and duration of the short-circuit should not exceed one second.

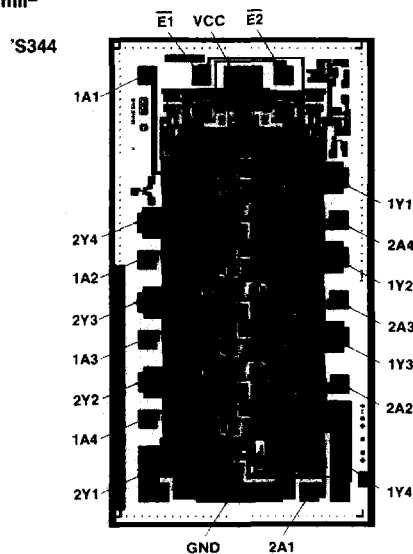
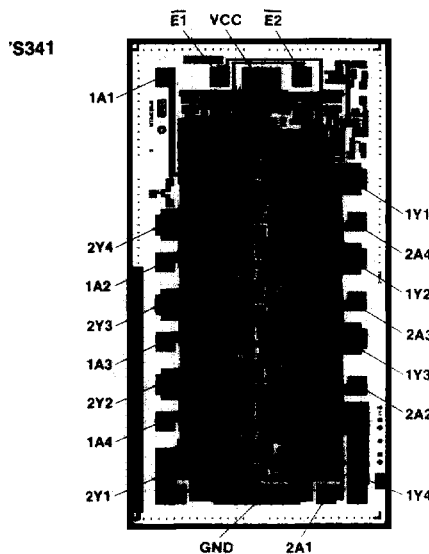
* "A" indicates data input, "E" indicates enable input.

Switching Characteristics $V_{CC} = 5\text{ V}$, $T_A = 25^\circ\text{C}$

SYMBOL	PARAMETER	TEST CONDITIONS (See Test Load/Waveforms)	'S310, 'S340 MIN TYP MAX		'S341, 'S344 MIN TYP MAX		UNIT
t _{PLH}	Data to Output delay	C _L = 50 pF R _L = 90 Ω	11	15	16	22	ns
t _{PHL}			16	22	10	15	ns
t _{PZL}	Output Enable delay		8	15	10	15	ns
t _{PZH}			6	12	7	12	ns
t _{PLZ}	Output Disable delay	C _L = 5 pF R _L = 90 Ω	10	15	10	15	ns
t _{PHZ}			7	12	7	12	ns

Die Configurations


Die Size: 62x121 mil²



Function Tables

'310

$\overline{E}_1$	E_2	1Y OUTPUTS	2Y OUTPUTS
H	H	Z	Enabled (Inverting)
H	L	Z	Z
L	H	Enabled (Inverting)	Enabled (Inverting)
L	L	Enabled (Inverting)	Z

'340

$\overline{E}_1$	$\overline{E}_2$	1Y OUTPUTS	2Y OUTPUTS
H	H	Z	Z
H	L	Z	Enabled (Inverting)
L	H	Enabled (Inverting)	Z
L	L	Enabled (Inverting)	Enabled (Inverting)

'341

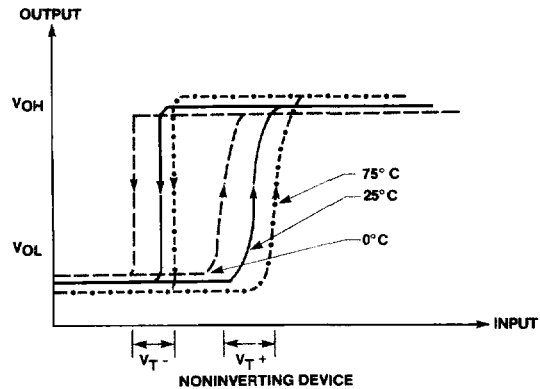
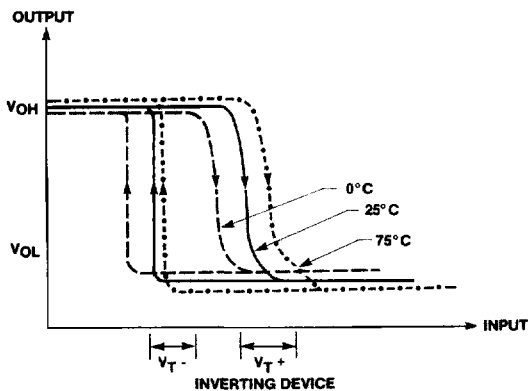
$\overline{E}_1$	E_2	1Y OUTPUTS	2Y OUTPUTS
H	H	Z	Enabled
H	L	Z	Z
L	H	Enabled	Enabled
L	L	Enabled	Z

'344

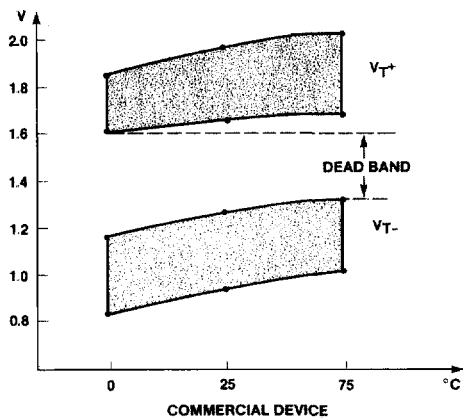
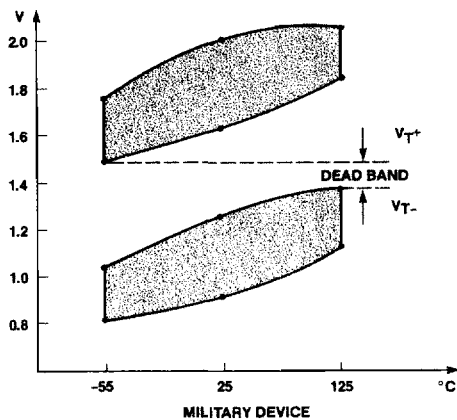
$\overline{E}_1$	$\overline{E}_2$	1Y OUTPUTS	2Y OUTPUTS
H	H	Z	Z
H	L	Z	Enabled
L	H	Enabled	Z
L	L	Enabled	Enabled

Z = High impedance (output off).

INPUT VS OUTPUT VOLTAGE TRANSFER CHARACTERISTIC

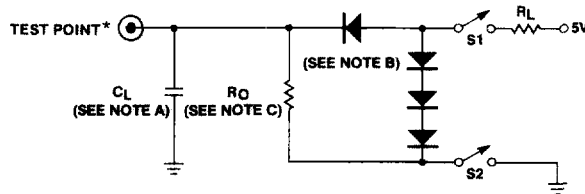


THRESHOLD VOLTAGE VS OPERATING TEMPERATURE



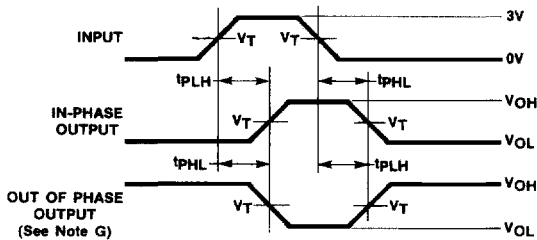
* Dead Band: The hysteresis is guaranteed at any operating temperature and voltage.

Test Load

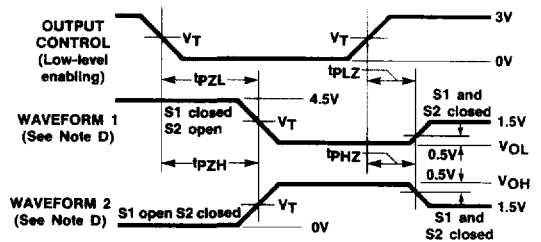


* The "TEST POINT" is driven by the output under test, and observed by instrumentation

Test Waveforms



Propagation Delay



Enable and Disable

- NOTES: A. C_L includes probe and jig capacitance.
 B. All diodes are 1N916 or 1N3064.
 C. For Series 54/74S310/340/341/344 $R_O = 5K$, $V_T = V_{T+} = 1.8V$ for low-to-high input transition.
 For Series 54/74S310/340/341/344 $R_O = 5K$, $V_T = V_{T-} = 1.1V$ for high-to-low input transition.
 For Series 54/74LS310/340/341/344 $R_O = 5K$, $V_T = V_{T+} = 1.7V$ for low-to-high input transition.
 For Series 54/74LS310/340/341/344 $R_O = 5K$, $V_T = V_{T-} = 0.9V$ for high-to-low input transition.
 D. Waveform 1 is for an output with internal conditions such that the output is low except when disabled by the output control.
 Waveform 2 is for an output with internal conditions such that the output is high except when disabled by the output control.
 E. In the examples above, the phase relationships between inputs and outputs have been chosen arbitrarily.
 F. All input pulses are supplied by generators having the following characteristics: $PRR \leq 1\text{ MHz}$, $Z_{OUT} = 50\Omega$ and:
 For Series 54/74S, $t_R \leq 2.5\text{ ns}$, $t_F \leq 2.5\text{ ns}$.
 For Series 54/74LS and PALs, $t_R \leq 15\text{ ns}$, $t_F \leq 6\text{ ns}$.
 G. When measuring propagation delay times of 3-state outputs, switches S1 and S2 are closed. (Propagation delays are measured from the inputs crossing V_{T+} , V_{T-} to the outputs crossing V_T)